SMART HOSTEL MANAGEMENT PLATFORM

Project Overview :

The **Smart Hostel Management Platform** is a multi-city hostel booking system designed to serve students, hostel owners, wardens, and platform administrators in Pakistani cities (Lahore, Karachi, Islamabad). The system supports flexible stay durations (daily, weekly, monthly, semester), paid amenities, meal subscriptions, automated invoicing, and comprehensive audit logging.

The system supports flexible stay durations (daily, weekly, monthly, semester) and enables real-time bed-level booking with automated validation to prevent overbooking. Students can filter hostels by city, gender, availability, pricing, and amenities. Paid amenities and meal subscriptions are managed with inventory control to ensure availability and accurate billing.

The financial module includes automated invoice generation, multi-method payment support (cash, card, bank transfer, Easypaisa, JazzCash), overdue tracking, and revenue reporting for hostel owners. Operational features include booking extensions, room transfers, early checkouts with refund calculation, complaint handling, maintenance tracking, guest approvals, and notification management.

The database schema is normalized up to Third Normal Form (3NF) to ensure data integrity and eliminate redundancy. Performance is optimized using strategic indexing and validated through query analysis, while triggers enforce business rules such as overbooking prevention and automated billing.

Overall, the platform provides a scalable, secure, and efficient solution for modern hostel management in a multi-city environment.

User Role Definitions :

STUDENT

End users who book hostel beds and consume services.

Permissions / Capabilities:

- Search and view available hostels and rooms
- Create, view, and cancel own bookings

- Subscribe to meal plans and paid amenities
- Make payments via multiple payment methods
- Submit maintenance requests and complaints
- Register guests for short-term visits
- View own invoices and payment history
- Extend bookings or request early checkout

Primary Tables Accessed:
bookings, payments, invoices, complaints, maintenance_requests, guest_registrations, student_meal_subscriptions, booking_amenities

WARDEN
Hostel staff responsible for day-to-day operations and student management.

Permissions / Capabilities:

- Approve or reject guest registrations
- Approve room transfer requests
- Manage and update maintenance requests
- Resolve complaints submitted by students
- View hostel occupancy and room status
- Approve booking extensions and early checkouts
- Receive high-turnover and maintenance notifications

Primary Tables Accessed:
maintenance_requests, complaints, guest_registrations, room_transfers, booking_extensions, early_checkouts, notifications

HOSTEL OWNER
Property owners who list and manage hostel properties on the platform.

Permissions / Capabilities:

- Create and manage hostel listings (rooms, beds, amenities)

- Set and override amenity prices
- Configure meal plans per hostel
- View revenue summaries and monthly earnings reports
- Define room categories and short-term policies
- Receive commission-based earnings statements
- Monitor occupancy dashboards

Primary Tables Accessed:
hostels, rooms, beds, amenities, hostel_amenities, meal_plans, room_categories, invoices (read)

SUPER ADMIN
Platform-level administrator with unrestricted access across all data.

Permissions / Capabilities:

- Manage all users, roles, and permissions
- Create and deactivate hostel owner accounts
- Configure commission rates per owner
- Access full audit logs and system activity
- Override any booking, payment, or complaint status
- Generate platform-wide analytics and reports
- Configure system-wide settings and parameters

Primary Tables Accessed:
ALL tables including audit_logs, roles, users, hostel_owners

Table Description :

1. **roles**
   Stores system-defined user roles such as student, warden, hostel_owner, and super_admin. It defines the permission structure and access level within the system.

2. **users**
   Stores all registered users of the platform including students, wardens, hostel owners, and administrators. Each user is linked to a specific role.

3. **hostel_owners**
   Stores additional business-related information for users who act as hostel owners, including company details, license information, bank account, and commission rate.

4. **hostels**
   Stores core information about each hostel such as name, city, address, geographic coordinates, gender restriction, total capacity, description, and rating.

5. **room_categories**
   Defines different stay duration categories such as daily, weekly, monthly, semester, or annual. It controls minimum and maximum stay rules.

6. **rooms**
   Stores details of rooms within each hostel including room number, floor, capacity, pricing, occupancy count, status, and category.

7. **beds**
   Represents individual beds inside rooms. Tracks bed type and availability status to allow precise booking management.

8. **amenities**
   Stores available optional services such as AC, WiFi, laundry, and gym, including pricing and classification category.

9. **hostel_amenities**
   Junction table linking hostels and amenities. Tracks availability count, custom pricing per hostel, and activation status.

10. **meal_plans**
    Stores meal packages offered by hostels including meal combinations (breakfast, lunch, dinner) and monthly pricing.

11. **bookings**
    Stores reservation records made by students. Includes booking dates, pricing, duration, status, and the bed assigned.

12. **booking_amenities**
    Junction table linking bookings with selected amenities. Stores amenity pricing and subscription duration for each booking.

13. **student_meal_subscriptions**
    Stores student subscriptions to meal plans including start date, end date, and subscription status.

14. **invoices**
    Stores billing information generated for bookings. Tracks amount due, amount paid, due date, late fees, and payment status.

15. **payments**
    Stores payment transactions made by users. Linked to invoices and bookings. Tracks payment method, amount, reference, and status.

16. **room_transfers**
    Stores requests and approvals for transferring a student from one room to another. Tracks approval details and status.

17. **booking_extensions**
    Stores booking extension requests including new end date, additional cost, and approval status.

18. **early_checkouts**
    Stores records of early checkout requests including refund amount, refund status, and reason.

19. **guest_registrations**
    Stores visitor records for students including guest details, check-in and check-out dates, approval status, and approver.

20. **maintenance_requests**
    Stores maintenance issues reported for rooms. Tracks priority level, status, and resolution timestamps.

21. **complaints**
    Stores formal complaints submitted by users about hostels. Includes category, priority, status, and resolution details.

22. **notifications**
    Stores system-generated alerts and messages sent to users regarding bookings, payments, maintenance, or general updates.

23. **audit_logs**
    Stores system activity logs for monitoring and auditing purposes. Tracks user actions, affected tables, previous and new values, and timestamps.

Data Dictionary:

Table: roles

Column Name	Data Type	Constraints	Description
role_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each role.
role_name	ENUM('student', 'warden', 'hostel_owner', 'super_admin')	NOT NULL, UNIQUE	Defines the type of system role.
permissions	JSON	None	Stores role-based permissions in JSON format.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp when role record was created.

Table: users

Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each user.
role_id	INT	NOT NULL, FOREIGN KEY → roles(role_id)	Links user to assigned role.
email	VARCHAR(255)	NOT NULL, UNIQUE	User login email.
password_hash	VARCHAR(255)	NOT NULL	Encrypted user password.
full_name	VARCHAR(100)	NOT NULL	Full name of user.

phone	VARCHAR(20)	None	Contact number.
city	VARCHAR(50)	None	City of residence.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp.

Table: hostel_owners

Column Name	Data Type	Constraints	Description
owner_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for hostel owner.
user_id	INT	NOT NULL, UNIQUE, FOREIGN KEY → users(user_id), ON DELETE CASCADE	Links owner record to user account.
company_name	VARCHAR(100)	None	Registered company name.
business_license	VARCHAR(50)	None	Official license number.
bank_account	VARCHAR(30)	None	Bank account for revenue transfer.
commission_rate	DECIMAL(5,2)	DEFAULT 10.00	Platform commission percentage.

created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp.

Table: hostels

Column Name	Data Type	Constraints	Description
hostel_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique hostel identifier.
owner_user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Owner of the hostel.
hostel_name	VARCHAR(100)	NOT NULL	Name of hostel.
city	VARCHAR(50)	NOT NULL	City location.
address	TEXT	None	Full address.
latitude	DECIMAL(10,8)	None	Geographic latitude.
longitude	DECIMAL(11,8)	None	Geographic longitude.
gender	ENUM('male', 'female', 'co-ed')	NOT NULL	Gender restriction policy.
total_capacity	INT	None	Total number of beds available.

description	TEXT	None	Hostel description.
rating	DECIMAL(3,2)	DEFAULT 0.00	Average user rating.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.

Table: room_categories

Column Name	Data Type	Constraints	Description
category_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for room stay category.
category_name	VARCHAR(50)	NOT NULL	Name of the stay category.
description	TEXT	None	Explanation of category purpose.
min_stay_days	INT	NOT NULL	Minimum allowed stay duration in days.
max_stay_days	INT	NOT NULL	Maximum allowed stay duration in days.
is_active	BOOLEAN	DEFAULT TRUE	Indicates whether category is currently active.

Table: rooms

Column Name	Data Type	Constraints	Description
room_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for room.
hostel_id	INT	NOT NULL, FOREIGN KEY → hostels(hostel_id), ON DELETE CASCADE	Hostel to which room belongs.
room_number	VARCHAR(10)	NOT NULL, UNIQUE (hostel_id, room_number)	Room number within hostel.
floor_number	INT	NOT NULL	Floor location of room.
capacity	INT	NOT NULL, CHECK (capacity > 0 AND capacity <= 6)	Maximum number of occupants.
occupied_count	INT	DEFAULT 0	Current number of occupied beds.
room_type	ENUM('single', 'double', 'triple', 'quad')	NOT NULL	Type of room based on bed count.
base_price_monthly	DECIMAL(10,2)	NOT NULL	Monthly rental price.
daily_rate	DECIMAL(10,2)	None	Daily rental rate (for short-term stays).
room_category_id	INT	FOREIGN KEY → room_categories(category_id)	Associated stay category.

allows_short_term	BOOLEAN	DEFAULT FALSE	Indicates if room allows short-term bookings.
status	ENUM('available', 'full', 'maintenance')	DEFAULT 'available'	Current room status.
has_attached_bathroom	BOOLEAN	DEFAULT FALSE	Whether room has attached bathroom.

Table: beds

Column Name	Data Type	Constraints	Description
bed_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique bed identifier.
room_id	INT	NOT NULL, FOREIGN KEY → rooms(room_id), ON DELETE CASCADE	Room containing the bed.
bed_number	INT	NOT NULL, UNIQUE (room_id, bed_number)	Bed number within room.
bed_type	ENUM('single', 'bunk_upper', 'bunk_lower')	DEFAULT 'single'	Type of bed.
status	ENUM('available', 'occupied', 'maintenance', 'reserved')	DEFAULT 'available'	Current bed availability status.

Table: amenities

Column Name	Data Type	Constraints	Description
amenity_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique amenity identifier.
amenity_name	VARCHAR(50)	NOT NULL, UNIQUE	Name of amenity.
description	TEXT	None	Description of amenity.
base_price_monthly	DECIMAL(10,2)	NOT NULL	Standard monthly cost.
category	ENUM('comfort', 'utility', 'service', 'entertainment')	NOT NULL	Amenity classification type.
icon_url	VARCHAR(255)	None	URL for amenity icon.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.

Table: hostel_amenities

Column Name	Data Type	Constraints	Description
hostel_amenity_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique record identifier.
hostel_id	INT	NOT NULL, FOREIGN KEY → hostels(hostel_id), ON DELETE CASCADE	Associated hostel.

amenity_id	INT	NOT NULL, FOREIGN KEY → amenities(amenity_id)	Linked amenity.
available_count	INT	NOT NULL	Quantity available.
price_override	DECIMAL(10,2)	None	Custom price for this hostel.
is_active	BOOLEAN	DEFAULT TRUE	Whether amenity is active at hostel.

Table: meal_plans

Column Name	Data Type	Constraints	Description
meal_plan_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique meal plan identifier.
hostel_id	INT	NOT NULL, FOREIGN KEY → hostels(hostel_id), ON DELETE CASCADE	Hostel offering the meal plan.
plan_name	VARCHAR(50)	NOT NULL	Name of meal plan.
description	TEXT	None	Description of included meals.
breakfast	BOOLEAN	DEFAULT FALSE	Includes breakfast.
lunch	BOOLEAN	DEFAULT FALSE	Includes lunch.

dinner	BOOLEAN	DEFAULT FALSE	Includes dinner.
price_monthly	DECIMAL(10,2)	NOT NULL	Monthly cost.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.

Table: bookings

Column Name	Data Type	Constraints	Description
booking_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking identifier.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Student who made the booking.
bed_id	INT	NOT NULL, FOREIGN KEY → beds(bed_id)	Assigned bed for booking.
booking_start_date	DATE	NOT NULL	Start date of stay.
booking_end_date	DATE	NOT NULL, CHECK (booking_start_date < booking_end_date)	End date of stay.
base_price	DECIMAL(10,2)	NOT NULL	Base cost of booking.
total_amenities_price	DECIMAL(10,2)	DEFAULT 0.00	Total cost of selected amenities.

stay_duration_days	INT	None	Total number of stay days.
stay_category	ENUM('daily', 'weekly', 'monthly', 'semester')	NOT NULL	Type of stay duration.
status	ENUM('confirmed', 'pending', 'cancelled', 'completed')	DEFAULT 'pending'	Current booking status.
payment_deadline	TIMESTAMP	None	Deadline for completing payment.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Booking creation timestamp.

Table: booking_amenities

Column Name	Data Type	Constraints	Description
booking_amenity_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique record identifier.
booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id), ON DELETE CASCADE	Related booking.
amenity_id	INT	NOT NULL, FOREIGN KEY → amenities(amenity_id)	Selected amenity.
monthly_price	DECIMAL(10,2)	NOT NULL	Price charged for amenity.

start_date	DATE	NOT NULL	Start date of amenity subscription.
end_date	DATE	None	End date of amenity subscription.

Table: student_meal_subscriptions

Column Name	Data Type	Constraints	Description
subscription_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique subscription identifier.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Student subscribing.
hostel_id	INT	NOT NULL, FOREIGN KEY → hostels(hostel_id)	Hostel providing meal plan.
meal_plan_id	INT	NOT NULL, FOREIGN KEY → meal_plans(meal_plan_id)	Selected meal plan.
start_date	DATE	NOT NULL	Subscription start date.
end_date	DATE	None	Subscription end date.
status	ENUM('active', 'paused', 'cancelled')	DEFAULT 'active'	Subscription status.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp.

Table: invoices

Column Name	Data Type	Constraints	Description
invoice_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique invoice identifier.
booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id)	Associated booking.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	User responsible for payment.
invoice_number	VARCHAR(50)	NOT NULL, UNIQUE	Unique invoice reference number.
amount_due	DECIMAL(10,2)	NOT NULL	Total payable amount.
amount_paid	DECIMAL(10,2)	DEFAULT 0.00	Amount already paid.
due_date	DATE	NOT NULL	Payment due date.
status	ENUM('pending', 'partially_paid', 'paid', 'overdue')	DEFAULT 'pending'	Invoice payment status.
late_fee	DECIMAL(10,2)	DEFAULT 0.00	Additional penalty fee.
generated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Invoice generation time.

Table: payments

Column Name	Data Type	Constraints	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment identifier.
invoice_id	INT	NOT NULL, FOREIGN KEY → invoices(invoice_id)	Linked invoice.
booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id)	Related booking.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	User making payment.
amount	DECIMAL(10,2)	NOT NULL, CHECK (amount >= 0)	Payment amount.
payment_method	ENUM('cash', 'card', 'bank_transfer', 'easypaisa', 'jazzcash')	NOT NULL	Method of payment.
transaction_reference	VARCHAR(100)	None	External transaction ID.
payment_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Payment timestamp.
status	ENUM('pending', 'completed', 'failed', 'refunded')	DEFAULT 'pending'	Payment processing status.

Table: room_transfers

Column Name	Data Type	Constraints	Description
transfer_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique transfer request ID.
old_booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id)	Original booking reference.
new_booking_id	INT	FOREIGN KEY → bookings(booking_id)	New booking after transfer.
student_user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Student requesting transfer.
reason	TEXT	None	Reason for transfer.
status	ENUM('pending', 'approved', 'rejected', 'completed')	DEFAULT 'pending'	Transfer request status.
requested_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Request timestamp.
approved_by	INT	FOREIGN KEY → users(user_id)	Warden approving transfer.
approved_at	TIMESTAMP	None	Approval timestamp.

Table: booking_extensions

Column Name	Data Type	Constraints	Description
extension_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking extension identifier.
booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id)	Booking being extended.
original_end_date	DATE	NOT NULL	Original checkout date.
new_end_date	DATE	NOT NULL	Requested new checkout date.
additional_days	INT	NOT NULL	Number of additional days requested.
additional_cost	DECIMAL(10,2)	NOT NULL	Extra cost for extension.
status	ENUM('pending', 'approved', 'rejected')	DEFAULT 'pending'	Extension approval status.
requested_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Request submission time.
approved_by	INT	FOREIGN KEY → users(user_id)	Warden/admin approving request.

Table: early_checkouts

Column Name	Data Type	Constraints	Description

checkout_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique early checkout identifier.
booking_id	INT	NOT NULL, FOREIGN KEY → bookings(booking_id)	Booking being terminated early.
original_end_date	DATE	NOT NULL	Scheduled checkout date.
actual_checkout_date	DATE	NOT NULL	Actual departure date.
days_unused	INT	NOT NULL	Number of unused days.
refund_amount	DECIMAL(10,2)	DEFAULT 0.00	Refund amount calculated.
refund_status	ENUM('pending', 'processed', 'no_refund')	DEFAULT 'pending'	Refund processing state.
reason	TEXT	None	Reason for early checkout.
requested_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Request timestamp.

Table: guest_registrations

Column Name	Data Type	Constraints	Description

guest_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique guest registration ID.
student_user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Student hosting the guest.
guest_name	VARCHAR(100)	NOT NULL	Guest's full name.
guest_phone	VARCHAR(20)	None	Guest contact number.
relationship	VARCHAR(50)	None	Relationship to student.
check_in_date	DATE	NOT NULL	Guest arrival date.
check_out_date	DATE	NOT NULL, CHECK (check_in_date < check_out_date)	Guest departure date.
status	ENUM('pending', 'approved', 'checked_in', 'checked_out', 'rejected')	DEFAULT 'pending'	Guest approval and visit status.
approved_by	INT	FOREIGN KEY → users(user_id)	Warden approving guest entry.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration creation timestamp.

Table: maintenance_requests

Column Name	Data Type	Constraints	Description
request_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique maintenance request ID.
room_id	INT	NOT NULL, FOREIGN KEY → rooms(room_id)	Room requiring maintenance.
reported_by	INT	NOT NULL, FOREIGN KEY → users(user_id)	User who reported issue.
issue_description	TEXT	NOT NULL	Detailed issue explanation.
priority	ENUM('low', 'medium', 'high', 'urgent')	DEFAULT 'medium'	Urgency level.
status	ENUM('open', 'in_progress', 'resolved', 'closed')	DEFAULT 'open'	Maintenance progress status.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Request creation time.
resolved_at	TIMESTAMP	None	Resolution completion time.

Table: complaints

Column Name	Data Type	Constraints	Description

complaint_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique complaint identifier.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	User submitting complaint.
hostel_id	INT	NOT NULL, FOREIGN KEY → hostels(hostel_id)	Hostel being complained about.
category	ENUM('cleanliness', 'noise', 'staff', 'maintenance', 'food', 'security', 'other')	NOT NULL	Complaint type classification.
subject	VARCHAR(200)	NOT NULL	Complaint title.
description	TEXT	NOT NULL	Detailed complaint content.
priority	ENUM('low', 'medium', 'high', 'urgent')	DEFAULT 'medium'	Urgency level.
status	ENUM('open', 'in_progress', 'resolved', 'closed')	DEFAULT 'open'	Complaint resolution status.
is_anonymous	BOOLEAN	DEFAULT FALSE	Whether complainant identity is hidden.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Submission timestamp.
resolved_at	TIMESTAMP	None	Resolution time.

resolved_by	INT	FOREIGN KEY → users(user_id)	User who resolved complaint.

Table: notifications

Column Name	Data Type	Constraints	Description
notification_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique notification identifier.
user_id	INT	NOT NULL, FOREIGN KEY → users(user_id)	Notification recipient.
title	VARCHAR(100)	NOT NULL	Notification title.
message	TEXT	NOT NULL	Notification message body.
type	ENUM('booking', 'payment', 'maintenance', 'complaint', 'general')	NOT NULL	Notification category.
is_read	BOOLEAN	DEFAULT FALSE	Indicates whether notification has been read.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Notification creation time.

Table: audit_logs

Column Name	Data Type	Constraints	Description
log_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique audit log identifier.
user_id	INT	FOREIGN KEY → users(user_id)	User who performed the action (nullable for system actions).
action	VARCHAR(100)	NOT NULL	Type of operation performed (INSERT, UPDATE, DELETE).
table_affected	VARCHAR(50)	NOT NULL	Name of table affected.
record_id	INT	None	Primary key of affected record.
old_value	JSON	None	Previous record state.
new_value	JSON	None	Updated record state.
ip_address	VARCHAR(45)	None	IP address of user.
timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Time when action occurred.

.

Index Specifications & Rationale:

#	Table	Index Name	Columns	Rationale

1	users	idx_email	email	Login authentication queries filter by email on every request. Without this index, a full table scan occurs for every login attempt.
2	users	idx_role	role_id	Frequent filtering of users by role (e.g., finding all wardens or students) is accelerated by this index.
3	hostels	idx_city	city	The primary search filter for students is city. This index eliminates full table scans on hostel searches.
4	hostels	idx_gender	gender	Gender filter is always applied alongside city in availability searches, reducing result sets efficiently.
5	hostels	idx_owner	owner_user_id	Owner dashboard queries fetch all hostels belonging to a specific owner.
6	rooms	idx_status	status	Availability queries filter rooms by status = available. Index prevents full scans on large room tables.
7	rooms	idx_floor	floor_number	Trigger logic (prevent_wrong_floor_allocation) and floor-based filtering both benefit from this index.
8	rooms	idx_hostel	hostel_id	JOIN between hostels and rooms on hostel_id is one of the

				most frequent operations in the system.
9	beds	idx_status	status	The overbooking prevention trigger and availability views both query beds by status on every booking attempt.
10	bookings	idx_user	user_id	Student booking history (student_booking_history view) always filters by user_id.
11	bookings	idx_dates	booking_start_date, booking_end_date	Date range queries for availability checking require efficient scanning of overlapping date ranges.
12	bookings	idx_status	status	Dashboard and reporting queries frequently filter bookings by status (confirmed, pending, etc.).
13	bookings	idx_booking_dates_range	booking_start_date, booking_end_date	Additional covering index for range-based availability queries with ORDER BY on dates.
14	bookings	idx_bookings_status_date	status, booking_start_date	Amenity revenue analysis queries filter by both status and start date simultaneously.
15	invoices	idx_user_status	user_id, status	Outstanding invoice lookups for a specific student filter on both columns simultaneously.
16	invoices	idx_due_date	due_date	Overdue invoice detection queries compare due_date <

				CURRENT_DATE on potentially millions of rows.
17	invoices	idx_invoices_status	status	Separate status index supports queries that filter only by payment status without user filter.
18	payments	idx_user	user_id	Payment history views per student require fast user-filtered access.
19	payments	idx_status	status	Revenue reporting filters completed payments; failed/refunded payments require separate tracking.
20	payments	idx_booking	booking_id	JOIN between payments and bookings in the revenue summary view uses this index heavily.
21	payments	idx_payment_date	payment_date	Monthly revenue aggregation in owner_revenue_summary groups by payment_date month.
22	amenities	idx_amenity_category	category	Amenity marketplace filtering by category (comfort, utility, etc.) uses this index.
23	maintenance_requests	idx_maintenance_status	status, priority	Maintenance queue sorted by priority within each status; composite index covers both filter columns.

24	notifications	idx_user_unread	user_id, is_read	Unread notification badge count (WHERE user_id = X AND is_read = 0) runs on every page load.
25	hostel_amenities	idx_hostel_amenities_active	hostel_id, is_active	Amenity display on hostel pages filters by hostel_id and is_active = TRUE simultaneously.